

What It Does, and Why

Features, design decisions, and how every requirement is met

Product — Trisheet: equity-research-grade company profiles for US-listed companies

Core claim — every financial figure traces back to an SEC filing

Requirements of record — the seven non-negotiable rules in `CLAUDE.md`

Date — 5 August 2026

What the product is

Trisheet takes a US-listed ticker and produces a company profile of the kind an equity research analyst would write — except that every single number in it can be traced, in one click, to the exact government filing it came from.

The problem it solves is a real one. Financial summaries are easy to generate and hard to trust. A language model asked about a company's revenue will produce a confident, plausible, well-formatted number — and no way to check it. For anything involving money, a figure you cannot verify is worse than no figure at all, because it invites a decision it cannot support.

The product's value is provenance, not automation. That sentence is the first line of the project's rules, and every architectural decision follows from it. Speed and polish are secondary. If a figure cannot name its source, it does not appear.

Why it is designed this way

The central problem: a model that sounds right

A language model is very good at writing and very bad at remembering numbers precisely. Asked for a company's operating margin it will produce something that looks correct, is formatted correctly, and may be wrong by a wide margin. It cannot tell you which it is.

The design response is to give the model no opportunity to be wrong about numbers at all:

THE MODEL IS NEVER ASKED TO...	INSTEAD
Calculate anything	All arithmetic happens in Python, in one file, using pandas and numpy
Recall a figure	Figures are extracted from the SEC's structured data and handed to it
Estimate a missing value	Missing data renders as "Not disclosed"
Decide what is trustworthy	Source tiers are enforced in code, before the model is called
Be trusted on output	Everything it writes is checked against the fact store afterwards

The model is enclosed on both sides. The gate closes before it runs; the audit runs after. Its job is reduced to the one thing it is genuinely good at — writing clear sentences about figures it was given.

Why provenance is a type, not a policy

Most systems that promise sourcing implement it as a convention: remember to attach the source. Conventions decay. So here, the central data structure — `Fact` — declares its five provenance fields with no default values. A fact that cannot name its source cannot be constructed. The rule is enforced by the shape of the data, not by anyone's diligence.

The database then declares the same five columns NOT NULL, so the rule holds a second time one layer further down, even if the code above were bypassed entirely.

Why exactly one file may touch market data

The rule that market prices must never appear in the audited financial statements section could have been a guideline. Instead it is physical: exactly one module in the codebase is permitted to import a market data provider. If any other file did, that is a bug, findable by search.

That single choke point makes the tier rule checkable in seconds rather than argued about in review. Three independent walls then stop a market figure reaching the financial highlights: the module tags its output, the write gate blocks the tier, and the verification gate checks again.

Why the report looks like paper

The report deliberately does not look like a dashboard. Financial documents are read on white paper: figures right-aligned in a monospaced typeface with tabular numerals so digits line up between rows, hairline rules instead of cards, no gradients, no shadows, no glow. The word "AI" appears nowhere in the interface — the reader is looking at sourced filings, and the machinery that assembled them is not the point.

The landing page is a deliberate exception. It is colourful and modern, because a first-time visitor needs an invitation before they need austerity. The split is intentional and documented, not an inconsistency.

What it can do — the feature list

Input and resolution

- Search by ticker or by company name, with live suggestions from the SEC's official index.
- Ambiguous searches return candidates rather than a guess — the user chooses.
- Automatic detection of US versus foreign filers, decided from the annual form the company actually filed, never assumed.
- Four analysis depths, controlling how many years are examined. Depth never relaxes a provenance rule.
- A reader may attach their own links — an investor relations page, a press release — which are quoted separately and can never become financial facts.

Data extraction

- Structured financial figures pulled from the SEC's XBRL data, with ordered tag fallbacks per metric because companies tag the same concept differently.
- US accounting taxonomy tried first, international taxonomy as fallback for foreign filers.
- Amendment precedence — a restated figure wins over the original it replaced.
- Segment breakdowns by business line.
- Narrative sections extracted as written: business description, risk factors, management discussion.
- Share price, market capitalisation and trading multiples, clearly marked as market data.
- A dated timeline of material recent filings.

Analysis

- Growth rates, margins and financial ratios across multiple years.
- Margin bridges showing what moved between periods.
- Discounted cash flow valuation, with sensitivity and scenario tables.
- Peer selection by a ladder of decreasing authority, recording which rung answered.
- Every derived value carries the formula that produced it and is labelled "calculated".

The report

- Seven sections: snapshot, business, financials, analysis, peers, developments, risks.
- **The provenance rail** — a persistent column of source cards showing form type, accession number, filing date and a direct link. Never collapsed into a footer.
- Every figure carries a superscript marker naming its source card. Hovering highlights it.
- A compliance strip showing fact count, tier distribution, citation coverage, and whether verification passed.
- Reconciliation statements — segments summing to consolidated revenue, assets equalling liabilities plus equity.
- Four charts: revenue and margin, cash flow, segment mix, peer valuation.
- Live progress feed while the report generates, showing each step and what it produced.

After the report

- A question-answering assistant that responds only from facts already carrying provenance, and redirects requests for investment advice rather than answering them.
- PDF export styled as an analyst tearsheet, with sources included.
- Excel export whose ratios are live formulas over the reported figures, not pasted values — so the reader can audit the arithmetic themselves.

How every requirement is met

The requirements of record for this project are the seven non-negotiable rules stated in `CLAUDE.md`. They are described there as architectural constraints, not preferences: violating any one is a defect regardless of whether tests pass. Each is addressed below with the mechanism that enforces it and the evidence that it holds.

Rule 1 — The model never performs arithmetic

Mechanism. All calculation lives in a single analysis module, written in pure functions with pandas and numpy — no file access, no global state, no side effects. The prose module builds its prompt from already-computed fact objects only.

Evidence. The analysis module is one of two carrying mandatory unit test coverage. Its purity means identical input yields byte-identical output, which is what makes a report reproducible.

Rule 2 — No fact exists without provenance

Mechanism. The `Fact` type declares tier, source type, source URL, accession number and filing date with no defaults. The write gate additionally rejects whitespace-only values that would satisfy the type while naming no source. The database declares all five columns NOT NULL.

Evidence. Verified in the source: all five fields present and required; the gate's blank-field check confirmed; every rejection logged rather than swallowed.

Rule 3 — Source tier enforcement is code, not prompting

Mechanism. The permitted tiers for the financial highlights section are a named constant. The write gate compares each fact's tier against it and refuses non-conforming facts with a typed rejection code and a logged reason.

Evidence. Confirmed in the source: the constant is set to Tiers 1 and 2 only, and the gate raises a section-blocked rejection for anything else. Enforced again independently at the verification gate.

Rule 4 — Exactly one module may import a market data provider

Mechanism. A single named module is the sole permitted importer. The rule is stated in that file's own header as a constraint on the rest of the codebase.

Evidence. A search across the entire backend for market provider imports returns matches in that module and nowhere else. **Verified clean.**

Rule 5 — Never fabricate, never estimate

Mechanism. A metric that resolves nowhere produces an explicit "not disclosed" marker rather than a gap or a zero. Derived values render with a "calculated" label and their formula. The verification gate rejects any number in the prose that does not exist in the fact store.

Evidence. The interface contains no occurrence of "N/A"; "Not disclosed" is used throughout. The "calculated" label was confirmed rendering on derived figures.

Rule 6 — Only SEC EDGAR is a hard dependency

Mechanism. The failure policy is stated once in the pipeline. Resolution, discovery and XBRL extraction may fail the report; every other step is allowed to produce nothing and settles as skipped or failed with a reason. Nothing raises a stack trace to the interface.

Evidence. Requesting a report ID that does not exist renders: *"No report with that id. It may have been generated by a server that has since restarted."* — what happened and what to do, in the interface's voice.

Rule 7 — Nothing is hardcoded to a single company

Mechanism. Filer type is detected from the most recent annual form; sector from the SEC's industry code; reporting currency from the units the company's own data uses; metric templates switch on sector. Every interface literal lives in one constants file.

Evidence. Live-tested against four tickers, including a foreign filer:

TICKER	COMPANY	FILER TYPE	CURRENCY	YEAR END	SECTOR FROM CODE
AAPL	Apple Inc.	domestic	USD	26 Sep	Electronic computers
MSFT	Microsoft Corp	domestic	USD	30 Jun	Prepackaged software
SHOP	Shopify Inc.	domestic	USD	31 Dec	Prepackaged software
TSM	Taiwan Semiconductor	foreign	TWD	31 Dec	Semiconductors

Three different financial year ends, two currencies, both filer types, four industry classifications — all detected, none hardcoded.

Engineering standards

The project sets explicit code standards. Their status as of 5 August 2026:

STANDARD	STATUS	DETAIL
Python fully type-hinted, type checker clean	Pass	31 source files, no errors
Python linting clean	Pass	All checks passed
TypeScript strict mode, no any	Pass	Compiles with no errors
Frontend linting clean	Pass	No warnings
Production build succeeds	Pass	12 of 12 pages generated
Unit tests on analysis and verification modules	Pass	457 tests, all passing
Runs against 3+ tickers including a foreign filer	Pass	4 tested, 1 foreign
Errors handled, nothing crashes	Pass	Unknown IDs render a stated message
No magic numbers — constants in config	Pass	217 named constants in one file
Structured JSON logging, correlated by report	Pass	Confirmed in running server output
Secrets from environment only, never logged	Pass	Wrapped so an accidental log cannot leak them
Conventional commit messages, one change each	Pass	Consistent across history

Deployment

COMPONENT	WHERE IT RUNS
Website	Vercel, deployed automatically from the main branch
Backend	Railway, deployed automatically from the main branch
Database and file storage	Supabase

Both halves are live and verified responding. The backend correctly restricts which web origins may call it, and the deployed website is on that list.

What the product deliberately does not do

The refusals are as much a part of the design as the features, and each is a decision rather than a gap:

IT WILL NOT...	BECAUSE
Give investment advice	The system reports what companies filed. It does not recommend, and the assistant redirects such questions.

Estimate a missing figure	An estimate presented beside sourced figures borrows their credibility without earning it.
Read headcount out of prose	A number parsed from a sentence is not a reported figure. Most companies never tag headcount, so it correctly shows "Not disclosed".
Guess which company you meant	A correct-looking report about the wrong business is the worst failure available to this product.
Let a web page become a financial fact	Attached links are quoted separately, marked "supplied" and "not verified", and cannot enter the financial highlights.
Hide that a figure failed verification	Blocking findings mark the report as failed and are shown beside the figures they concern. The reader is told, rather than shown a clean page.

The one sentence that summarises the whole project. Everything here is arranged so that the interesting claim — *every figure traces to a filing* — is true because it cannot be false, rather than true because everyone was careful.